NLP Course Project Proposal · Track B: Demonstration
Team: Captain Marvel

1. Project Title

MomentJump: Natural Language Semantic Retrieval and Timestamp Localization for General Web Videos. The system covers three content regimes—**speech-centric**, **dialogue-centric**, and **event-centric** videos—whose primary signals are speech (ASR), dialogue subtitles, and commentary plus discrete events, respectively.

2. Team Members & Responsibilities

Member / ID	Role	Responsibilities
Zhang Shuhan / 2413157	Lead · Retrieval Core	Hybrid retrieval (Dense+BM25+RRF), Cross-Encoder reranking, query rewriting; system architecture
Teng Xin / 2414166	Ingest Pipeline	ffmpeg audio extraction, faster-whisper ASR, chunking (fixed/sliding/pause-boundary), multi-signal indexing
Li Zifei / 2312183	Data & Evaluation	Corpus curation, query→gold timestamp annotation, metrics (R@K / mIoU / temporal error), unified ablations
Fu Zixuan / 2313715	System & Demo	FastAPI backend, Streamlit UI, domain switching, player seek, deployment & demo recording
Zhao Yuejiao / 2313650	Domain Adapters · Report	SRT/scene blocks, event-log alignment, failure analysis, README / report / slides

3. Background & Motivation

Web videos are information-dense yet hard to navigate: users seeking "the few seconds on a concept," "a specific line," or "a goal moment" often rely on titles, chapter markers, or scrubbing. Keyword search and platform chapters operate at title/episode granularity and rarely support second-level localization. The ACL 2025 Demo **LECTURE4ALL** shows that ASR transcription, semantic chunking, and dense retrieval enable precise timestamping for lecture videos, but evaluation focuses on usability (SUS), **lacks standard IR metrics**, and centers on English, single-domain content.

Our motivation is to treat lectures, drama, and sports not as isolated applications but as three content-signal regimes, and to build a **unified text-segment retrieval framework** evaluated with standard IR metrics across all three. We will implement **chunking strategies, hybrid retrieval, query rewriting, Cross-Encoder reranking, and an automatic evaluation suite**, yielding a demonstrable, general, and measurable system.

4. Problem Definition / Research Questions

Task. Given a natural-language query and a video corpus, return the most relevant clips `(video_id, start_sec, end_sec, snippet)` for one-click playback.
Input. Chinese/English queries (colloquial questions, technical phrases, plot descriptions, event descriptions).
Output. A Top-K ranked list with timestamps, matched text, confidence, and play links.
Success criterion. Predicted span IoU $\geq \theta$ with gold, or predicted center within gold $\pm \delta$ (δ = 10s / 15s).

Generality. All domains map to a unified `Segment` (`domain` $\in$ {speech, dialogue, event}). The retrieval stack is domain-agnostic; only ingest signal sources differ (ASR / subtitles / commentary+events). Scope covers videos with transcribable text signals; pure visual, textless content is out of core scope.

- **RQ1 (Generality / effectiveness):** Does unified Hybrid (Dense+BM25+RRF) consistently outperform Dense-only and BM25-only across the three domains?
- **RQ2 (Segmentation):** How do fixed, sliding, and pause/punctuation chunking affect localization precision and recall? Does the best strategy differ by domain?
- **RQ3 (Signal availability):** How do speech / dialogue / commentary+event signals affect retrieval, and where does the framework fail?
- **RQ4 (Optional gains):** What gains do query rewriting and Cross-Encoder reranking bring in each domain?

5. System Design & Technical Approach

5.1 System Functions

1. Domain switching (speech / dialogue / event) with NL search returning Top-K clips.

2. Result cards with title, timecode, highlighted transcript, and similarity; click-to-seek from `start_sec`.
3. Side-by-side ablation panel (disable Hybrid / disable rerank) to visualize module gains.

5.2 Technical Architecture

Pipeline: video/subtitles/commentary → **Ingest** (ffmpeg → faster-whisper ASR / SRT parsing / event-log alignment → chunking → BGE-M3 embedding) → **Index** (FAISS + BM25 + SQLite metadata) → **Retrieval** (query rewrite → Hybrid → rerank → aggregation) → **Service** (FastAPI) → **UI** (Streamlit + HTML5 seek). Domains share a common `Segment` schema; retrieval is category-agnostic, with domain differences isolated in ingest.

Stack: faster-whisper (Chinese ASR); BGE-M3 / bge-base-zh (embeddings); bge-reranker-base; FAISS + rank_bm25; FastAPI + Streamlit.

5.3 Implementation Plan

In-house modules: (i) Chinese chunking ablations (fixed / sliding / pause–punctuation); (ii) Hybrid retrieval with RRF fusion; (iii) colloquial query rewriting; (iv) Cross-Encoder reranking and comparisons; (v) unified evaluation sets and automated metrics across three domains.

Reusable components: faster-whisper (ASR API only), pretrained BGE embedding/reranker weights, FAISS/rank_bm25, Streamlit/FastAPI frameworks—with clear reuse boundaries in the README.

Data (rights-cleared): speech-centric open lectures / self-recorded talks (3–10 h); dialogue-centric short dramas or public samples with custom SRT (2–5 clips); event-centric short highlights with 20–40 annotated events (incl. esports commentary). **40–80** `query→gold timestamp` items per domain.

5.4 Evaluation Plan

Metrics: Recall@1/@3/@5; mIoU; median absolute start-time error. Hit rule: IoU or center within $\delta = 10$s / 15s.

Baselines / ablations (identical suite on all three domains): BM25-only → Dense-only → Hybrid → Hybrid+Rerank; plus chunking and w/ vs w/o query rewrite. Cross-domain consistency of gains serves as evidence of generality.

6. Expected Outcomes

1. Runnable system + Web Demo (README with environment, commands, sample queries): domain switch, search → Top-K → seek-to-play.
2. Experimental report: unified R@K / temporal error / mIoU, ablations, generality claims, and failure cases.
3. Source code and configs (`requirements.txt`, `configs/default.yaml`).
4. **Target:** Hybrid+Rerank outperforms Dense-only and BM25-only on R@K and temporal error across domains; end-to-end reproducibility and live demo.

7. Project Timeline

Phase	Milestone	Tasks
Oct 2026	Spec freeze	Data & rights clearance, environment, v0 eval sets, architecture freeze
Mid-Nov 2026	MVP (midterm-ready)	≥3 speech videos transcribed & indexed; Dense/BM25 + Web seek; initial R@K; domain-agnostic retrieval & eval scripts
End of term	Full delivery	Dialogue/event adapters; full Hybrid+Rerank+rewrite ablations; report, slides, demo video

References

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5. *SoccerNet-Echoes.* arXiv:2405.07354.
6. Chen et al. *BGE-M3 (FlagEmbedding).* https://github.com/FlagOpen/FlagEmbedding
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